SG No. 124 D_{4h}^2 $P4/mcc$ [tetragonal]

* plus set: $+[0, 0, 0]$

Table 1: Wyckoff site: 2a, site symmetry: 422

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	[1,2,3,4,5,6,7,8]
2	$\left[0, 0, \frac{3}{4}\right]$	[9,10,11,12,13,14,15,16]

Table 2: Wyckoff site: 2b, site symmetry: 4/m..

No.	position	mapping
1	$[0, 0, 0]$	[1,2,3,4,9,10,11,12]
2	$\left[0, 0, \frac{1}{2}\right]$	[5,6,7,8,13,14,15,16]

Table 3: Wyckoff site: 2c, site symmetry: 422

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	[1,2,3,4,5,6,7,8]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	[9,10,11,12,13,14,15,16]

Table 4: Wyckoff site: 2d, site symmetry: 4/m..

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[1,2,3,4,9,10,11,12]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[5,6,7,8,13,14,15,16]

Table 5: Wyckoff site: 4e, site symmetry: 2/m..

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	[1,2,9,10]
2	$\left[\frac{1}{2}, 0, 0\right]$	[3,4,11,12]
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[5,6,13,14]
4	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[7,8,15,16]

Table 6: Wyckoff site: 4f, site symmetry: 222.

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[1,2,5,6]
2	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[3,4,7,8]
3	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[9,10,13,14]
4	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[11,12,15,16]

Table 7: Wyckoff site: `4g`, site symmetry: `4..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,4]`
2	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[5,6,7,8]`
3	$[0,\ 0,\ -z]$	`[9,10,11,12]`
4	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[13,14,15,16]`

Table 8: Wyckoff site: `4h`, site symmetry: `4..`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[5,6,7,8]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[9,10,11,12]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[13,14,15,16]`

Table 9: Wyckoff site: `8i`, site symmetry: `2..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,2]`
2	$\left[\frac{1}{2},\ 0,\ z\right]$	`[3,4]`
3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[5,6]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[7,8]`
5	$\left[0,\ \frac{1}{2},\ -z\right]$	`[9,10]`
6	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[11,12]`
7	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[13,14]`
8	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[15,16]`

Table 10: Wyckoff site: `8j`, site symmetry: `..2`

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{4}\right]$	`[1,7]`
2	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[2,8]`
3	$\left[-x,\ x,\ \frac{1}{4}\right]$	`[3,5]`
4	$\left[x,\ -x,\ \frac{1}{4}\right]$	`[4,6]`
5	$\left[-x,\ -x,\ \frac{3}{4}\right]$	`[9,15]`
6	$\left[x,\ x,\ \frac{3}{4}\right]$	`[10,16]`
7	$\left[x,\ -x,\ \frac{3}{4}\right]$	`[11,13]`
8	$\left[-x,\ x,\ \frac{3}{4}\right]$	`[12,14]`

Table 11: Wyckoff site: `8k`, site symmetry: `.2.`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,6]`
2	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[2,5]`

continued ...

Table 11

No.	position	mapping
3	$\left[0,\, x,\, \frac{1}{4}\right]$	[3,7]
4	$\left[0,\, -x,\, \frac{1}{4}\right]$	[4,8]
5	$\left[-x,\, 0,\, \frac{3}{4}\right]$	[9,14]
6	$\left[x,\, 0,\, \frac{3}{4}\right]$	[10,13]
7	$\left[0,\, -x,\, \frac{3}{4}\right]$	[11,15]
8	$\left[0,\, x,\, \frac{3}{4}\right]$	[12,16]

Table 12: Wyckoff site: 8l, site symmetry: .2.

No.	position	mapping
1	$\left[x,\, \frac{1}{2},\, \frac{1}{4}\right]$	[1,6]
2	$\left[-x,\, \frac{1}{2},\, \frac{1}{4}\right]$	[2,5]
3	$\left[\frac{1}{2},\, x,\, \frac{1}{4}\right]$	[3,7]
4	$\left[\frac{1}{2},\, -x,\, \frac{1}{4}\right]$	[4,8]
5	$\left[-x,\, \frac{1}{2},\, \frac{3}{4}\right]$	[9,14]
6	$\left[x,\, \frac{1}{2},\, \frac{3}{4}\right]$	[10,13]
7	$\left[\frac{1}{2},\, -x,\, \frac{3}{4}\right]$	[11,15]
8	$\left[\frac{1}{2},\, x,\, \frac{3}{4}\right]$	[12,16]

Table 13: Wyckoff site: 8m, site symmetry: m..

No.	position	mapping
1	$[x,\, y,\, 0]$	[1,10]
2	$[-x,\, -y,\, 0]$	[2,9]
3	$[-y,\, x,\, 0]$	[3,12]
4	$[y,\, -x,\, 0]$	[4,11]
5	$\left[-x,\, y,\, \frac{1}{2}\right]$	[5,14]
6	$\left[x,\, -y,\, \frac{1}{2}\right]$	[6,13]
7	$\left[y,\, x,\, \frac{1}{2}\right]$	[7,16]
8	$\left[-y,\, -x,\, \frac{1}{2}\right]$	[8,15]

Table 14: Wyckoff site: 16n, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$[-x,\, -y,\, z]$	[2]
3	$[-y,\, x,\, z]$	[3]
4	$[y,\, -x,\, z]$	[4]
5	$\left[-x,\, y,\, \frac{1}{2} - z\right]$	[5]
6	$\left[x,\, -y,\, \frac{1}{2} - z\right]$	[6]
7	$\left[y,\, x,\, \frac{1}{2} - z\right]$	[7]
8	$\left[-y,\, -x,\, \frac{1}{2} - z\right]$	[8]
9	$[-x,\, -y,\, -z]$	[9]

continued ...

Table 14

No.	position	mapping
10	$[x,\, y,\, -z]$	[10]
11	$[y,\, -x,\, -z]$	[11]
12	$[-y,\, x,\, -z]$	[12]
13	$\left[x,\, -y,\, z+\frac{1}{2}\right]$	[13]
14	$\left[-x,\, y,\, z+\frac{1}{2}\right]$	[14]
15	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	[15]
16	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[16]